Supplementary Information

Table I. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for the Tomato crop.

Model	**Classification**			**Segmentation**		**#Epoch**	**Total Loss**	**Weight (TFLite) [MB]**	**Parameters [M]**	**Inference [ms]**
	Accuracy [%]	**Recall [%]**	**F1-score [%]**	**Mean IoU [%]**	**Accuracy [%]**					
U-Net & ResNet 50	94.64	94.64	94.64	40.05	90.18	27	0.5039	18.11	32.37	298.00
U-Net & MobileNetV2	95.83	95.83	95.78	16.27	81.36	48	0.4797	11.94	3.15	151.07
Mask R-CNN & ResNet50	55.36	55.36	48.11	32.10	81.86	6	3.2282	117.14	30.80	94.40
Mask R-CNN & MobileNetV2	92.86	92.86	92.75	54.90	87.27	7	2.6227	24.84	6.55	67.41
DeepLabV3 & ResNet50	94.64	94.64	94.53	67.70	91.11	12	0.5793	119.97	31.55	146.32
DeepLabV3 & MobileNetV2	98.21	98.21	98.21	66.03	90.20	22	0.4461	20.79	5.49	87.98

Table II. Weighted scoring matrix for model selection in Tomato crop

Model	**Score Perf.(40%)**	**Score Latency(30%)**	**Score Weight(20%)**	**Score Param(10%)**	**Final Punctuation**
U-Net & ResNet 50	6.9 (2.76)	0	9.4 (1.88)	0	4.64
U-Net & MobileNetV2	4.67(1.887)	6.37 (1.9)	10 (2)	10 (1)	6.79
Mask R-CNN & ResNet50	1.5 (0.6)	8.83 (2.65)	0.26 (0.052)	0.68 (0.068)	3.37
Mask R-CNN & MobileNetV2	8.125 (3.25)	10 (3)	8.74 (1.748)	8.84 (0.88)	8.878
DeepLabV3 & ResNet50	9.585 (3.834)	6.58 (1.974)	0	0.28 (0.028)	5.84
DeepLabV3 & MobileNet	**9.84 (3.936)**	**9.1 (2.73)**	**9.1 (1.82)**	**9.2 (0.92)**	**9.41**

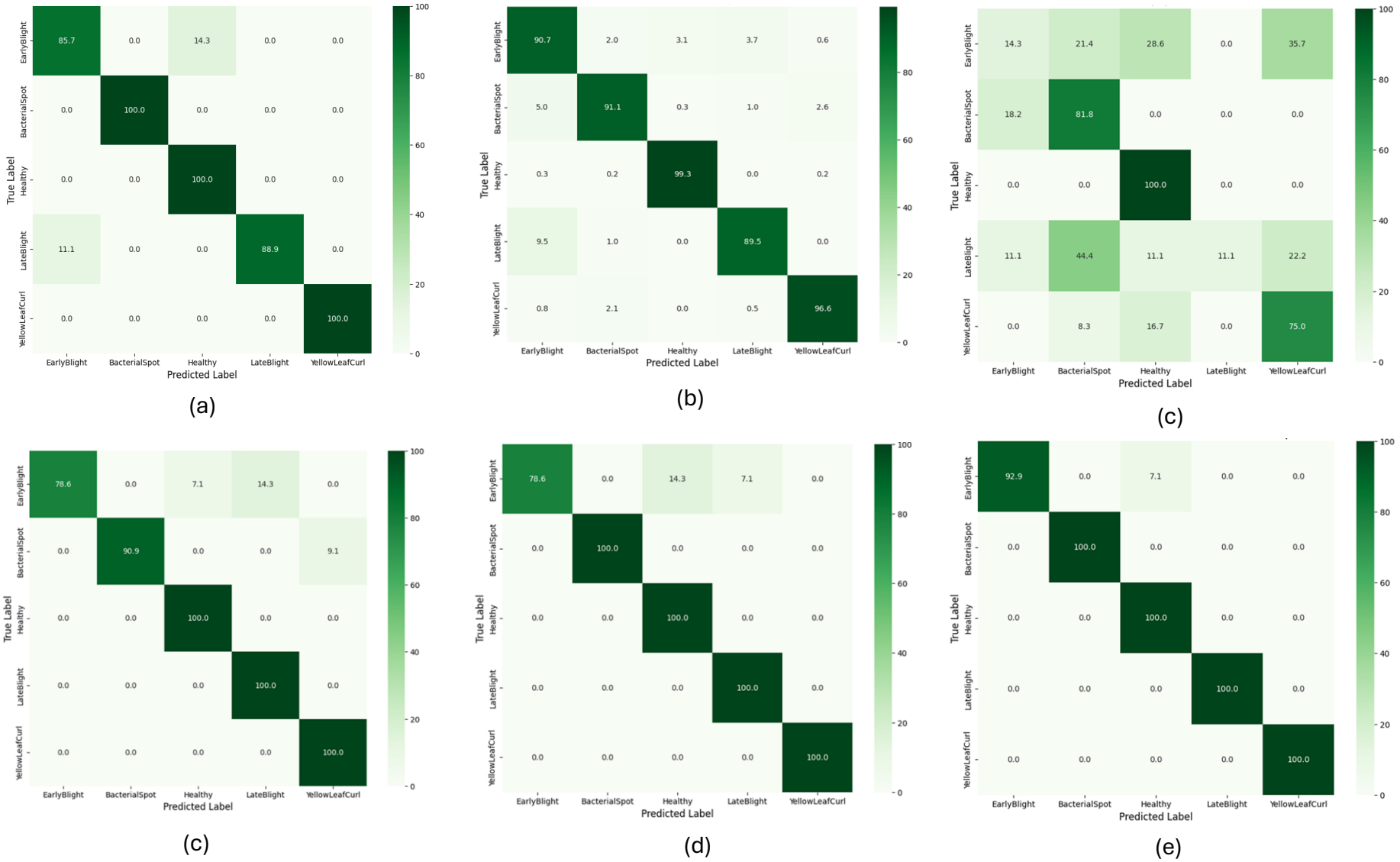


Fig. 1. Confusion matrix model of models in Tomato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

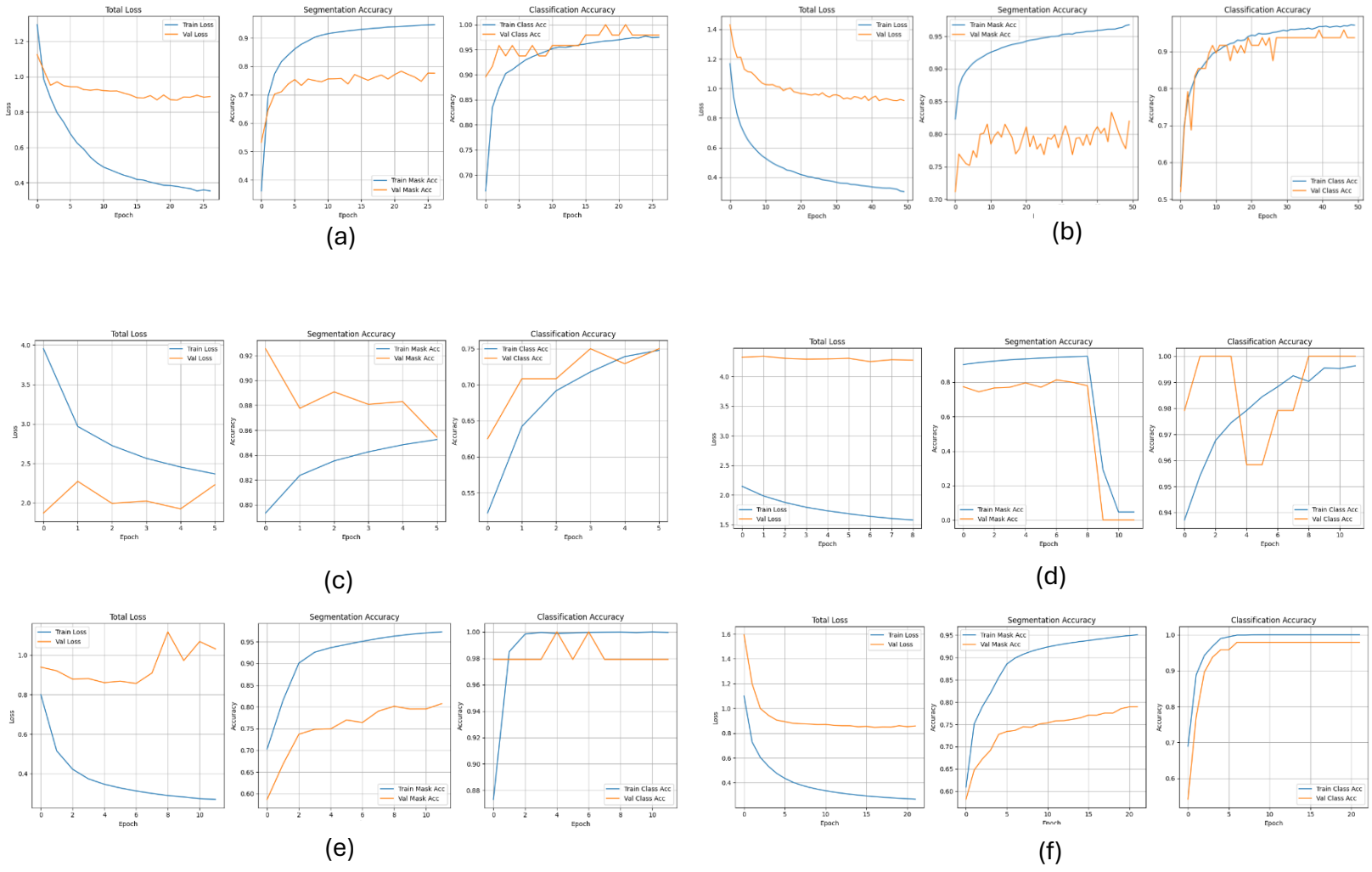


Fig.2. Loss curves in Tomato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

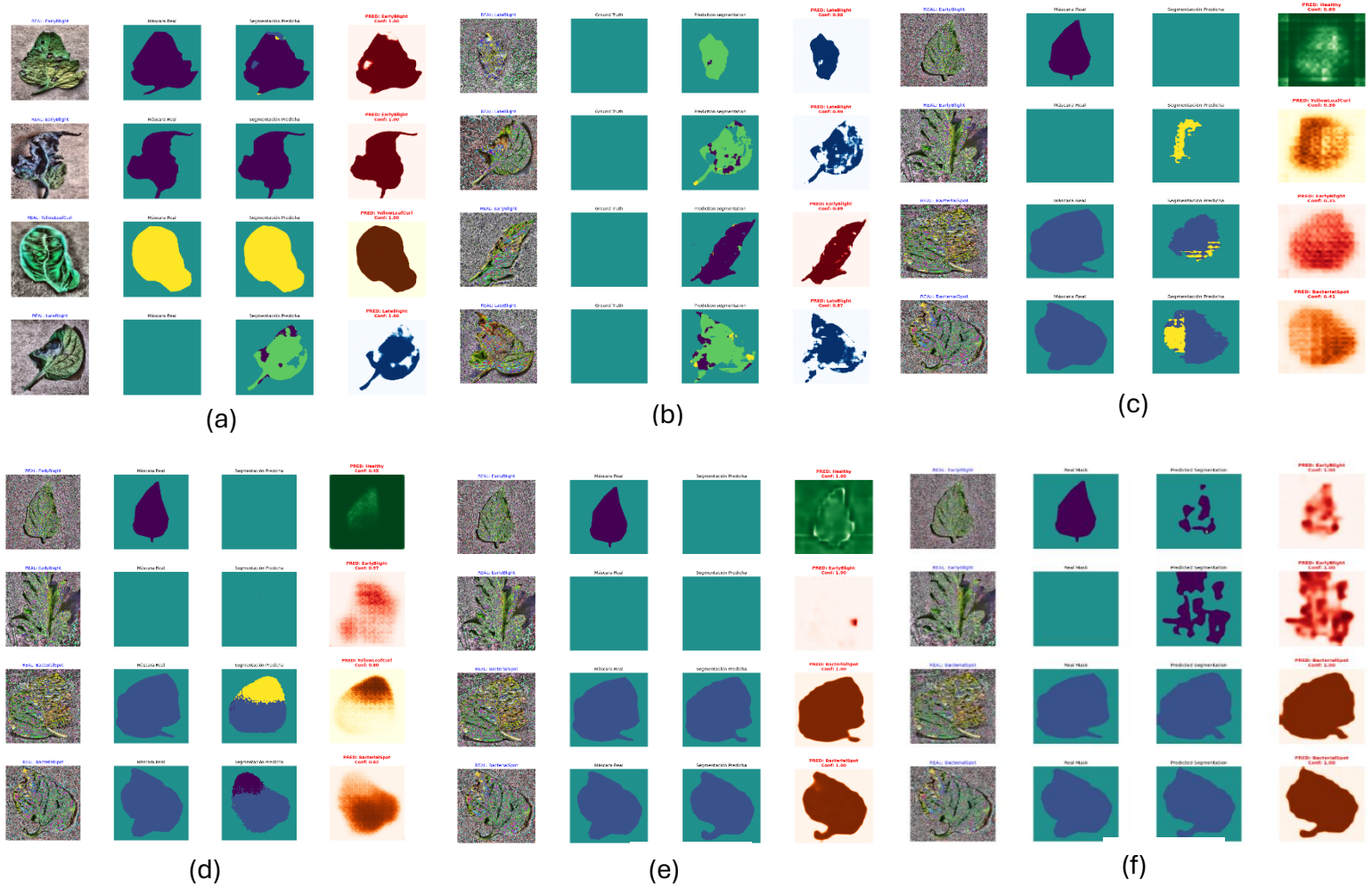


Fig. 3. Segmentation Samples for Tomato Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table III. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for Potato crop

Model	**Classification**			**Segmentation**		**#Epoch**	**Total Loss**	**Weigth (TFLite) [MB]**	**Parameters [M]**	**Inference [ms]**
	Accuracy [%]	**Recall [%]**	**F1-score [%]**	**Mean IoU [%]**	**Accuracy [%]**					
U-Net & ResNet50	96.43	96.43	96.27	34.27	76.76	21	0.6115	119.86	31.52	214.40
U-Net& MobileNetV2	98.21	98.21	98.19	81.89	91.33	13	0.2255	13.1	3.32	68.72
Mask R-CNN & ResNet50	83.93	83.93	83.56	49.25	76.43	13	0.5983	102.75	27.03	188.03
Mask R-CNN & MobileNetV2	96.43	96.43	96.27	35.61	71.22	14	0.7926	22.37	5.91	84.99
DeepLabV3+ & ResNet50	94.64	94.64	94.57	58.90	80.09	11	0.4641	110.65	29.10	514.75
DeepLabV3+ & MobileNetV2	98.21	98.21	98.19	42.73	44.35	14	0.9978	18.11	4.79	123.02

Table IV. Weighted scoring matrix for model selection in Potato crop

Model	**Score Perf.(40%)**	**Score Latency(30%)**	**Score Weight(20%)**	**Score Param(10%)**	**Final Punctuation**
U-Net & ResNet 50	4.38 (1.752)	5.15 (1.545)	0	0	3.297
U-Net & MobileNetV2	**10 (4)**	**10 (3)**	**10 (2)**	**10 (1)**	**10**
Mask R-CNN & ResNet50	1.575 (0.63)	7.32 (2.196)	1.6 (0.32)	1.59 (0.16)	3.31
Mask R-CNN & MobileNetV2	4.52 (1.808)	9.6 (2.88)	9.13 (1.826)	9.08 (0.908)	7.4
DeepLabV3 & ResNet50	4.24 (1.696)	0	0.863 (0.173)	0.86 (0.086)	1.955
DeepLabV3 & MobileNet	5.89 (2.356)	8.78 (2.634)	9.45 (1.89)	9.48 (0.948)	7.828

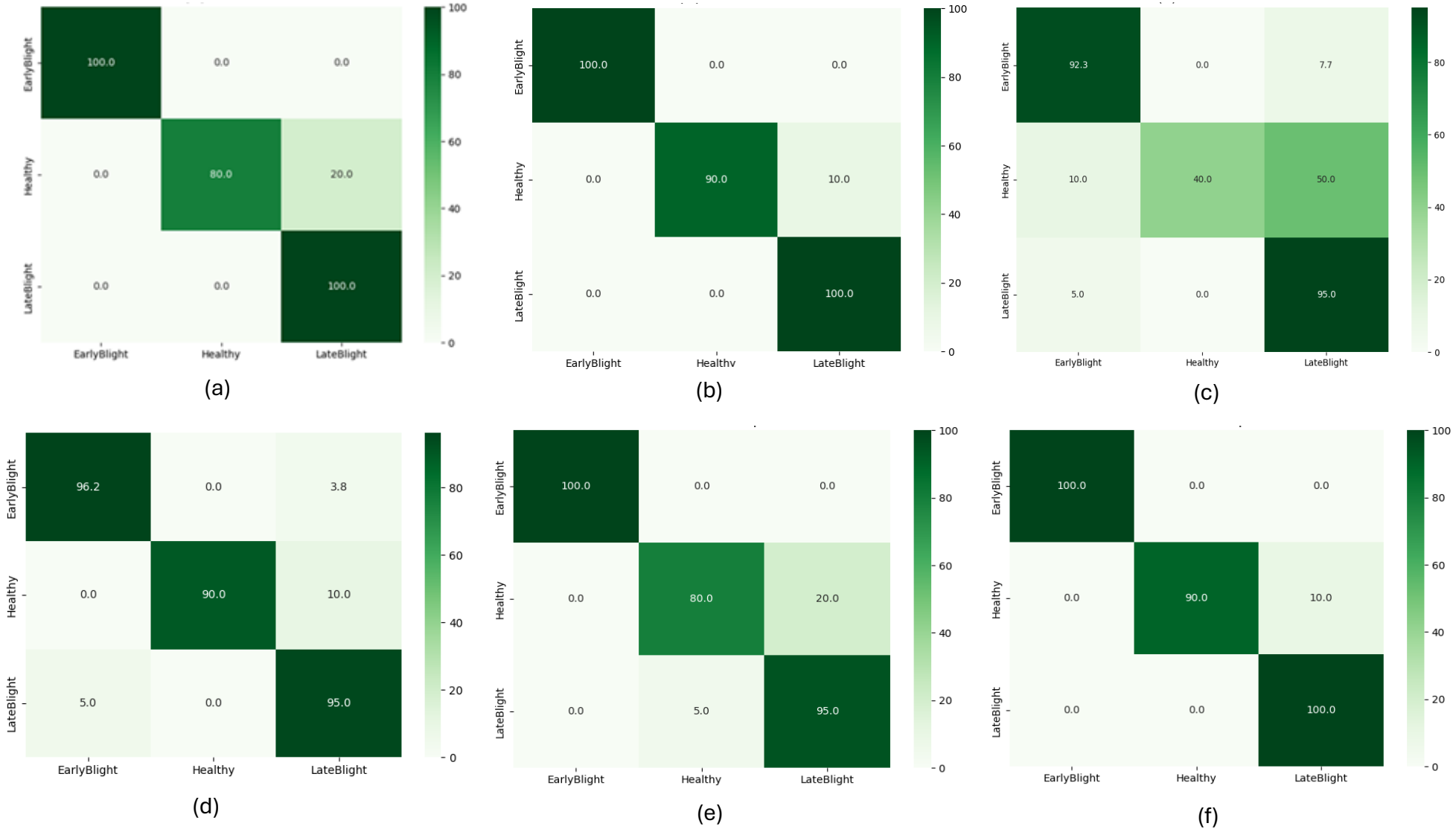


Fig. 4. Confusion matrix model of models in Potato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2.

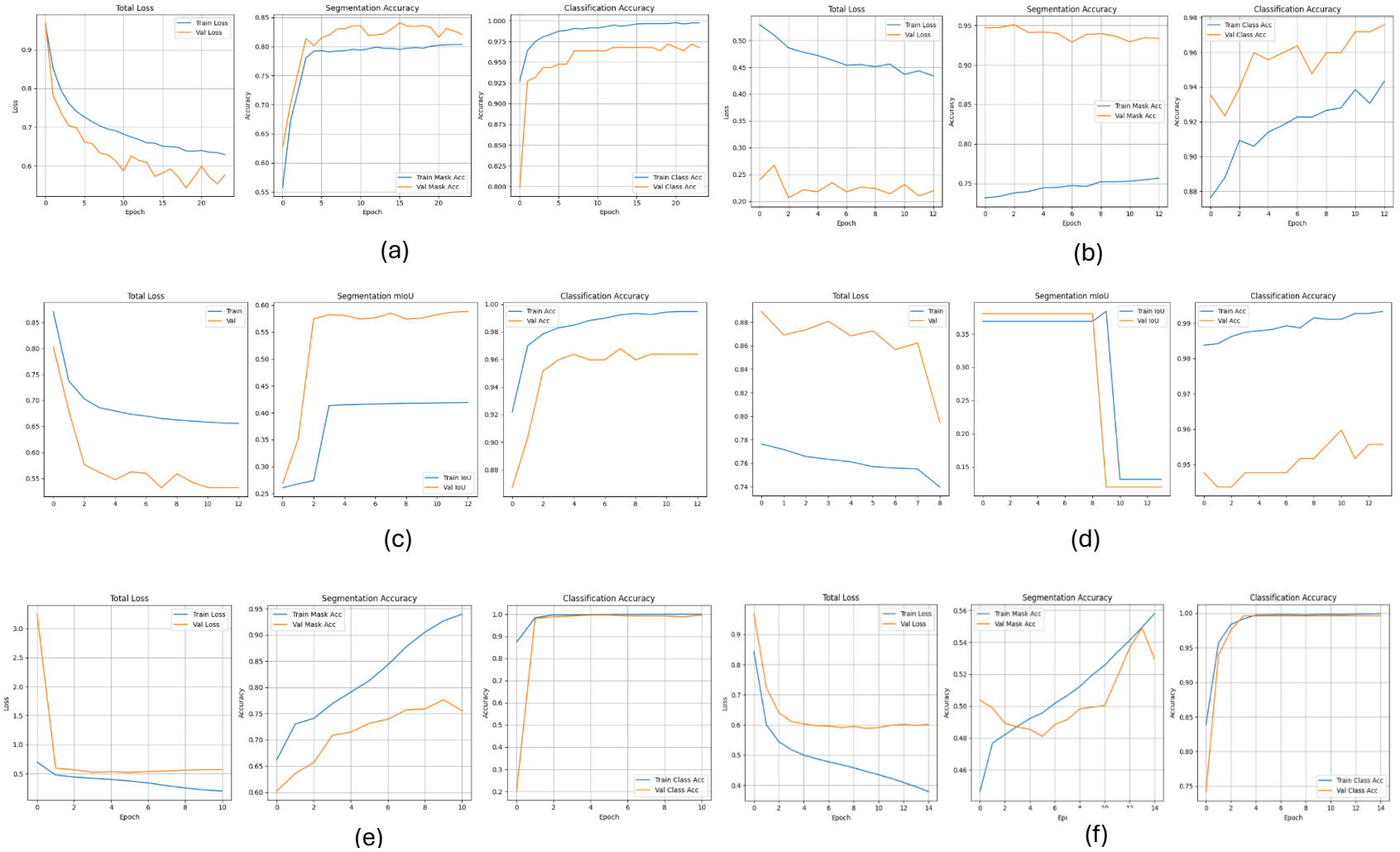


Fig.5. Loss curves in Potato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

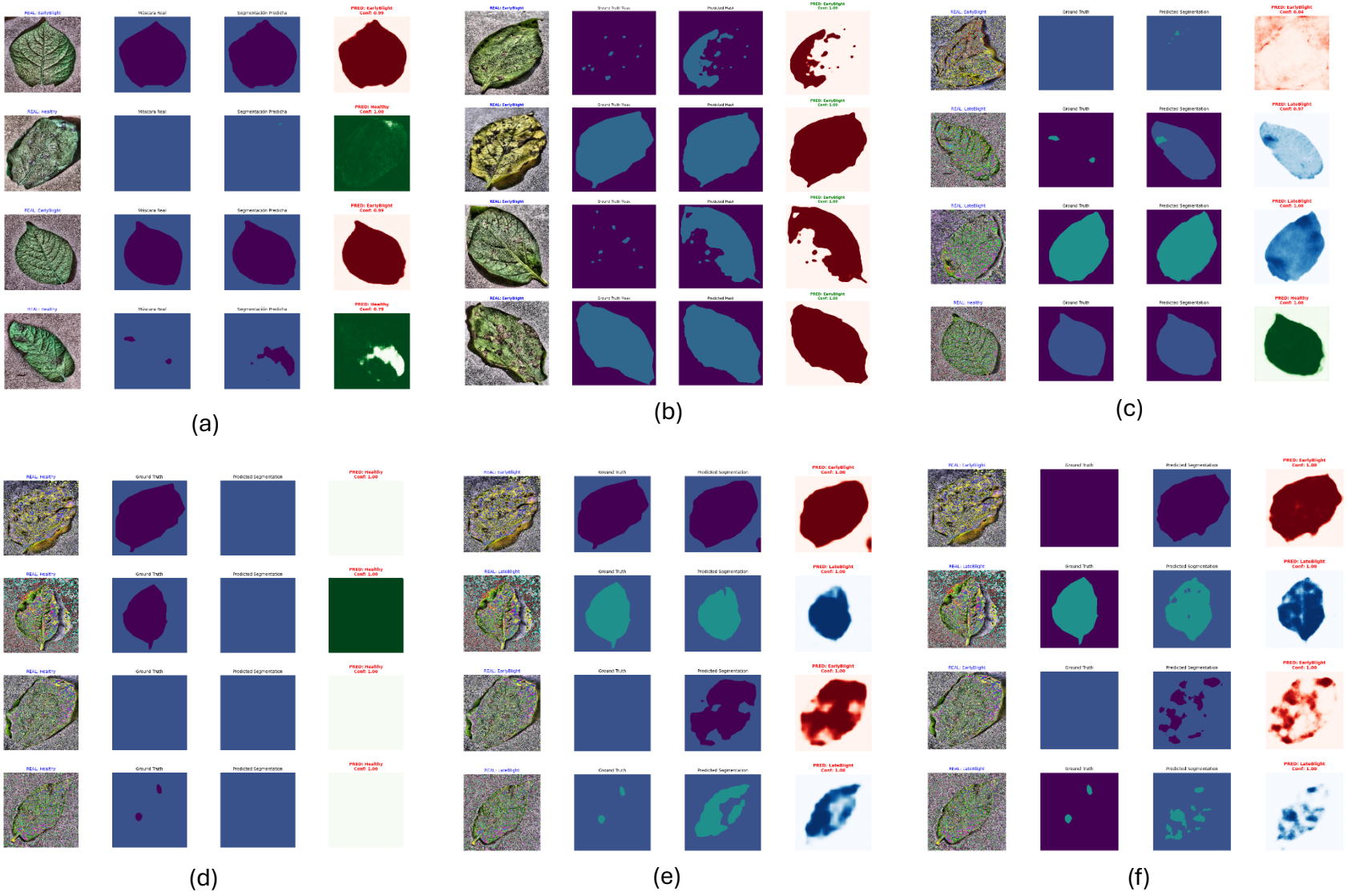


Fig. 6. Segmentation Samples for Potato Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table V. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for Bell Pepper crop

Model	Classification			Segmentation		#Epoch	Total Loss	Weigth (TFLite) [MB]	Parameters [M]	Inference [ms]
	Accuracy [%]	Recall [%]	F1-score [%]	Mean IoU [%]	Accuracy [%]					
U-Net & ResNet50	100	100	100	97.87	99.14	16	0.0213	5.05	1.34	130.52
U-Net& MobileNetV2	91.67	91.67	91.55	97.04	98.73	13	0.0562	13.38	3.56	87.05
Mask R-CNN & ResNet50	100	100	100	98.27	99.33	11	0.0293	91.18	23.94	105.65
Mask R-CNN & MobileNetV2	100	100	100	96.59	98.42	16	0.0408	11.23	32.76	87.65
DeepLabV3 & ResNet50	100	100	100	98.71	99.50	9	0.0357	40.33	10.59	141.61
DeepLabV3 & MobileNetV2	**100**	**100**	**100**	**98.91**	**99.63**	**38**	**0.0234**	**9.88**	**2.63**	**59.81**

Table VI. Weighted scoring matrix for model selection in Bell Pepper crop

Model	Score Perf. (40%)	Score Latency (30%)	Score Weight (20%)	Score Param (10%)	Final Punctuation
U-Net & ResNet 50	7.925 (3.17)	1.36 (0.41)	10 (2)	10 (1)	6.58
U-Net & MobileNetV2	0.835 (0.334)	6.61 (1.983)	9.03 (1.806)	9.3 (0.93)	5.053
Mask R-CNN & ResNet50	8.67 (3.468)	4.397 (1.32)	7.81 (1.562)	2.8 (0.28)	6.63
Mask R-CNN & MobileNetV2	5 (2)	6.597 (1.98)	9.28 (1.856)	0	5.836
DeepLabV3 & ResNet50	8.925 (3.57)	0	5.9 (1.18)	7.14 (0.714)	5.464
DeepLabV3 & MobileNet	10 (4)	10 (3)	9.44 (1.888)	9.595 (0.96)	9.848

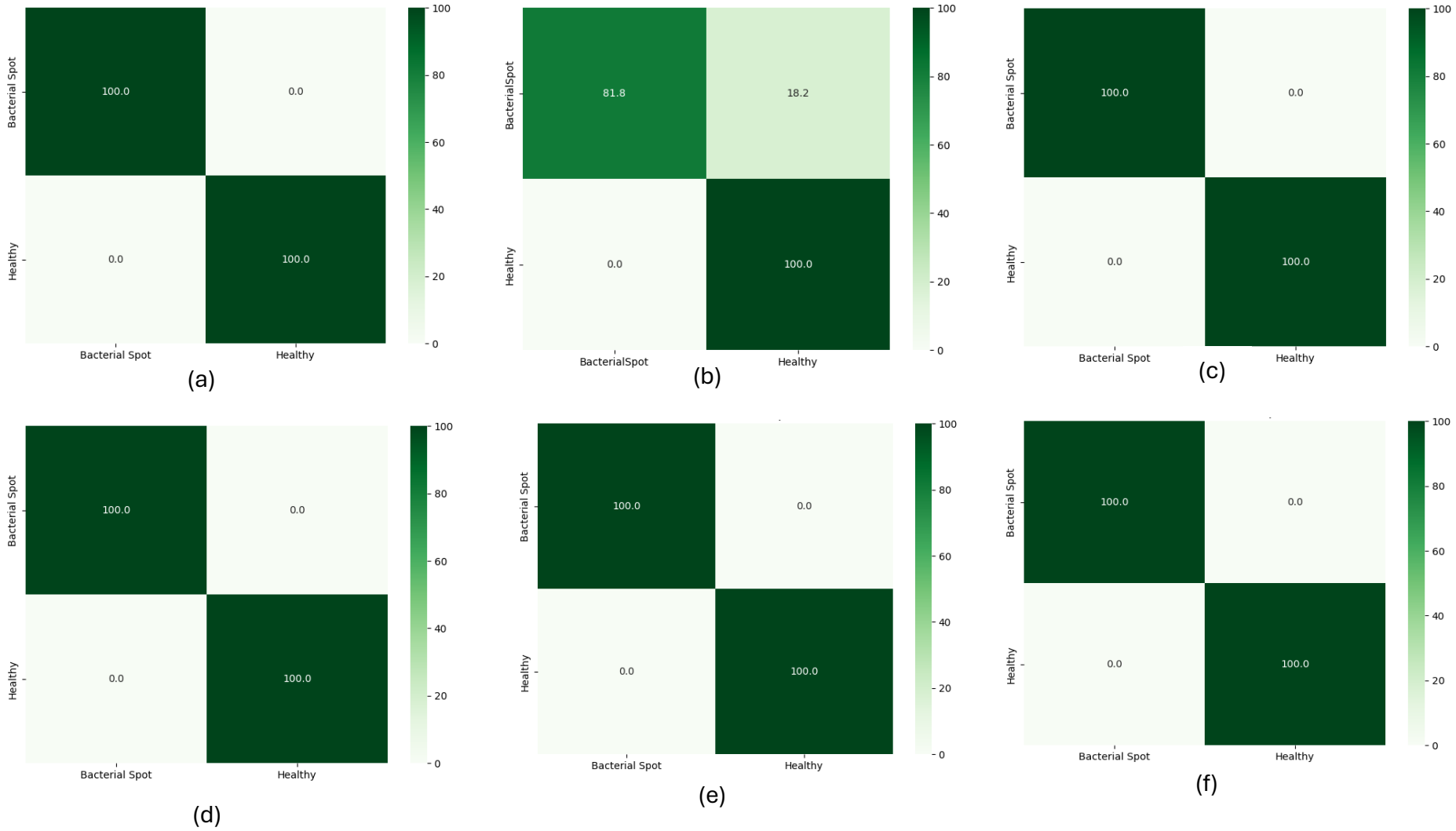


Fig. 7. Confusion matrix model of models in Bell Pepper crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

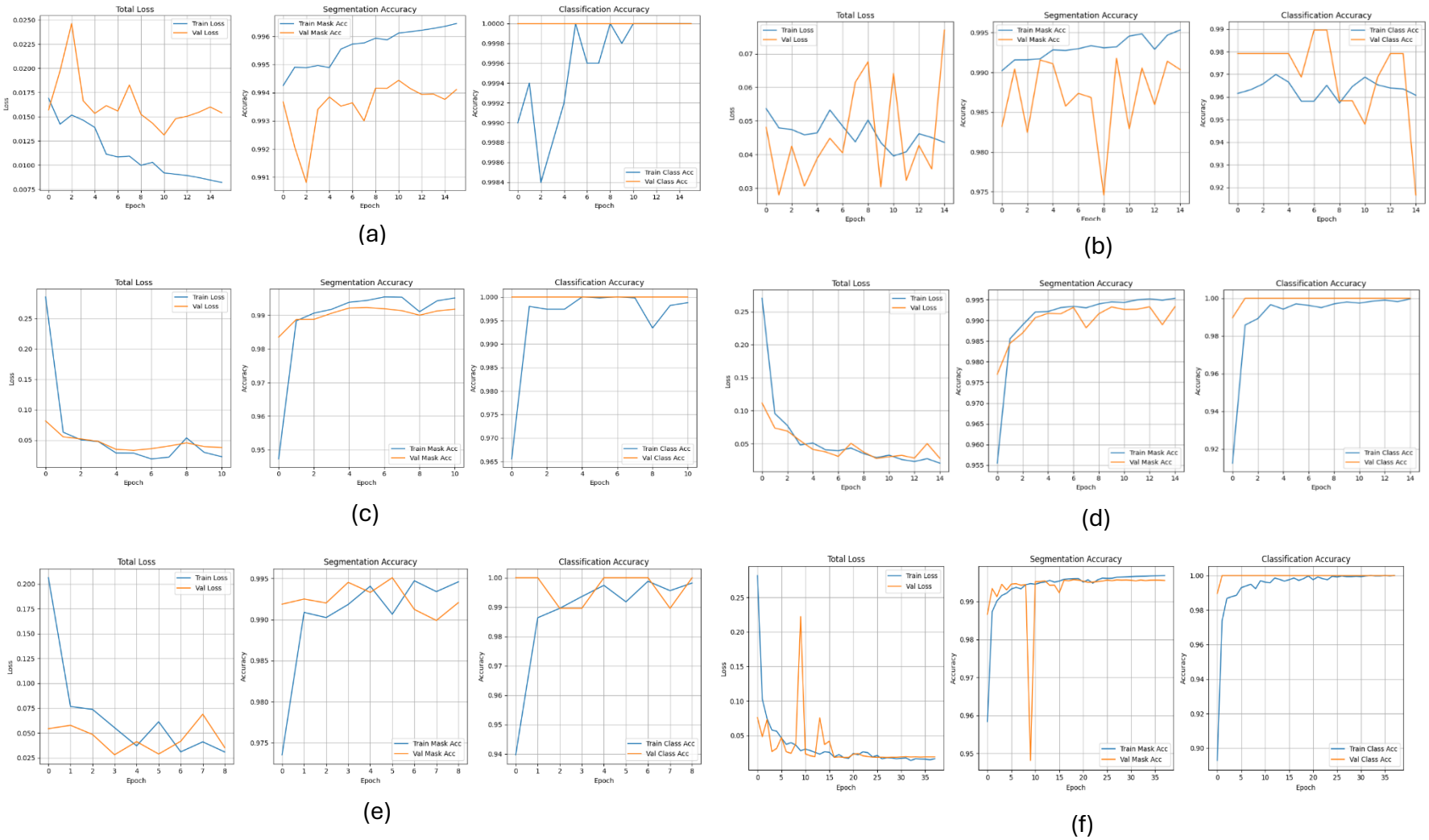


Fig.8. Loss curves in Bell Pepper crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

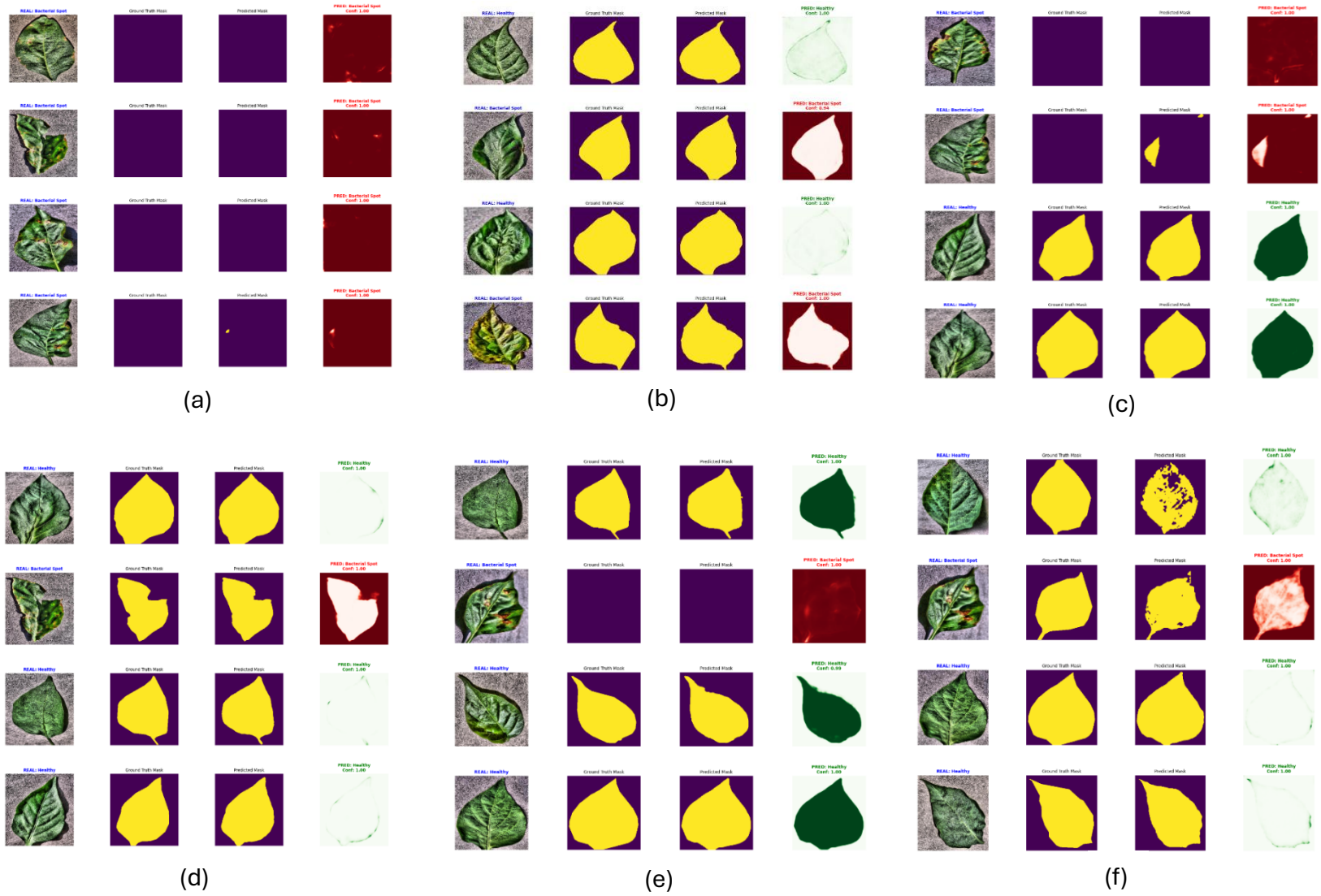


Fig. 9. Segmentation Samples for Bell Pepper Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table VII. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for Eggplant crop

Model	Classification			Segmentation		#Epoch	Total Loss	Weigth (TFLite) [MB]	Parameters [M]	Inference [ms]
	Accuracy [%]	Recall [%]	F1-score [%]	Mean IoU [%]	Accuracy [%]					
U-Net & ResNet 50	97.50	100	100	82.11	94.33	23	1.1344	167.46	43.99	267.65
U-Net& MobileNetv2	97.50	95	94.96	78.52	91.89	33	1.5599	13.41	3.55	77.73
Mask R-CNN & ResNet50	97.50	97.50	97.49	78.99	93.01	13	1.1926	123.98	32.59	128.57
Mask R-CNN & MobileNetv2	97.50	97.50	97.49	75.53	90.17	18	1.1097	12.62	3.35	66.59
DeepLabV3 & ResNet50	97.50	97.50	97.49	77.55	92.23	10	1.1479	115.65	30.41	137.12
DeepLabV3 & MobileNetv2	97.50	97.50	97.49	81.69	93.51	21	0.5465	9.89	2.63	66.99

Table VIII. Weighted scoring matrix for model selection in Eggplant crop

Model	Score Perf. (40%)	Score Latency (30%)	Score Weight (20%)	Score Param (10%)	Final Punctuation
U-Net & ResNet 50	10 (4)	0	0	0	4
U-Net & MobileNetV2	7.25 (2.9)	9.42 (2.826)	9.78 (1.956)	9.77 (0.977)	8.67
Mask R-CNN & ResNet50	7.63 (3.052)	6.93 (2.079)	2.76 (0.552)	2.76 (0.276)	5.96
Mask R-CNN & MobileNetV2	5 (2)	10 (3)	9.83 (1.966)	10 (1)	7.97
DeepLabV3 & ResNet50	6.535 (2.614)	6.74 (2.022)	3.288 (0.658)	3.283 (0.33)	5.62
DeepLabV3 & MobileNet	9.68 (3.872)	9.98 (2.994)	10 (2)	10 (1)	9.87

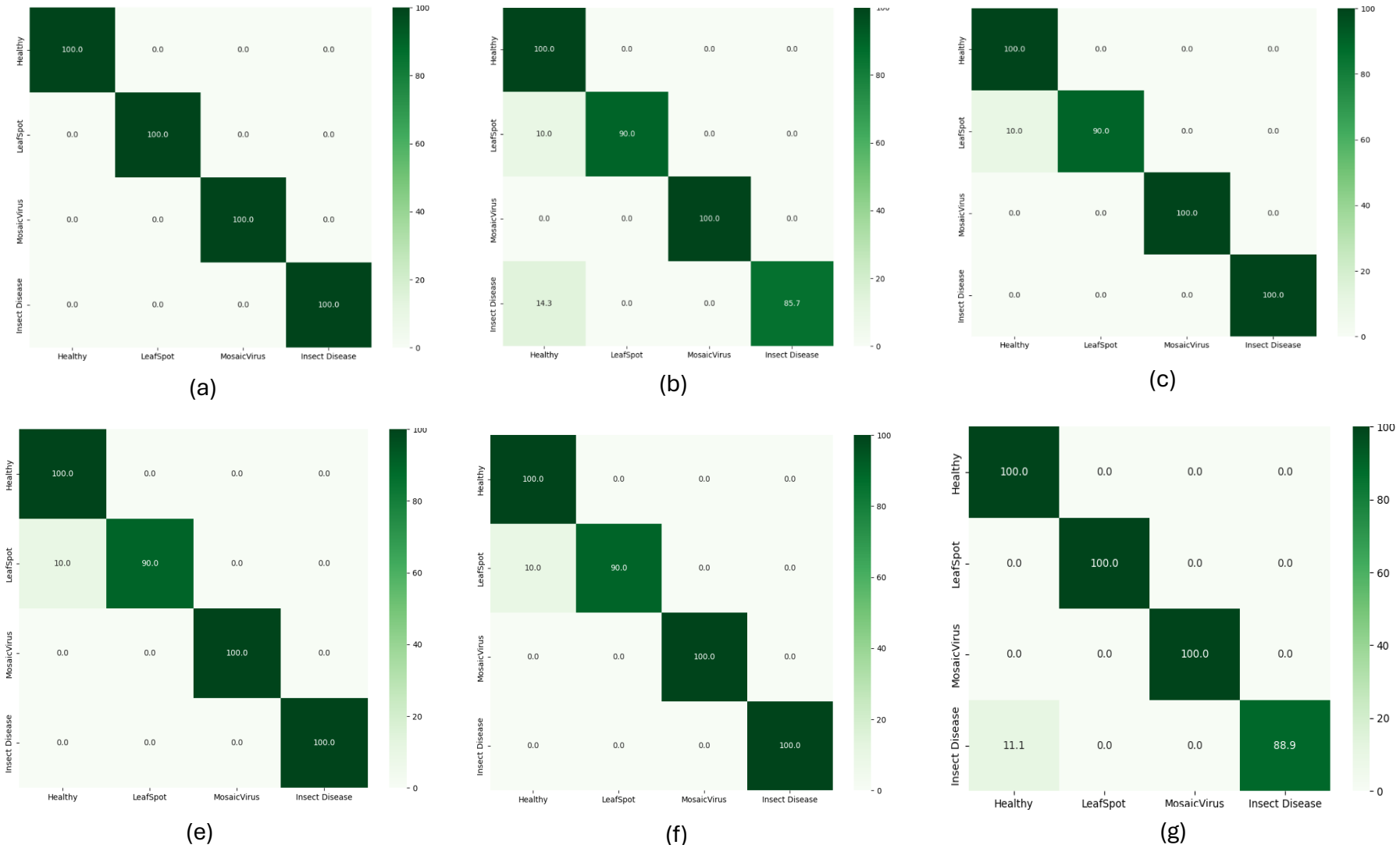


Fig. 10. Confusion matrix model of models in Eggplant crop. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2.

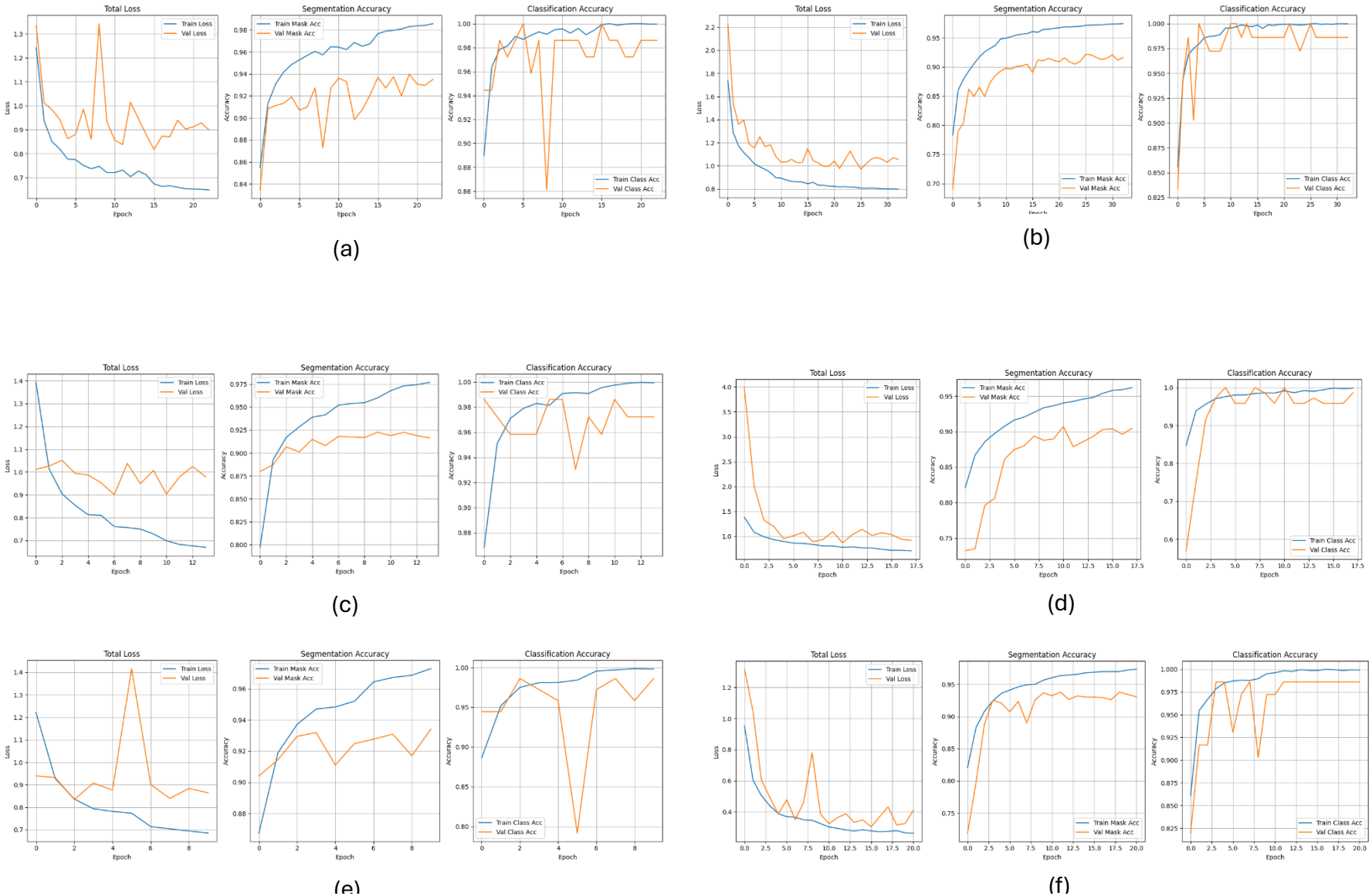


Fig.11. Loss curves in Eggplant crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2.

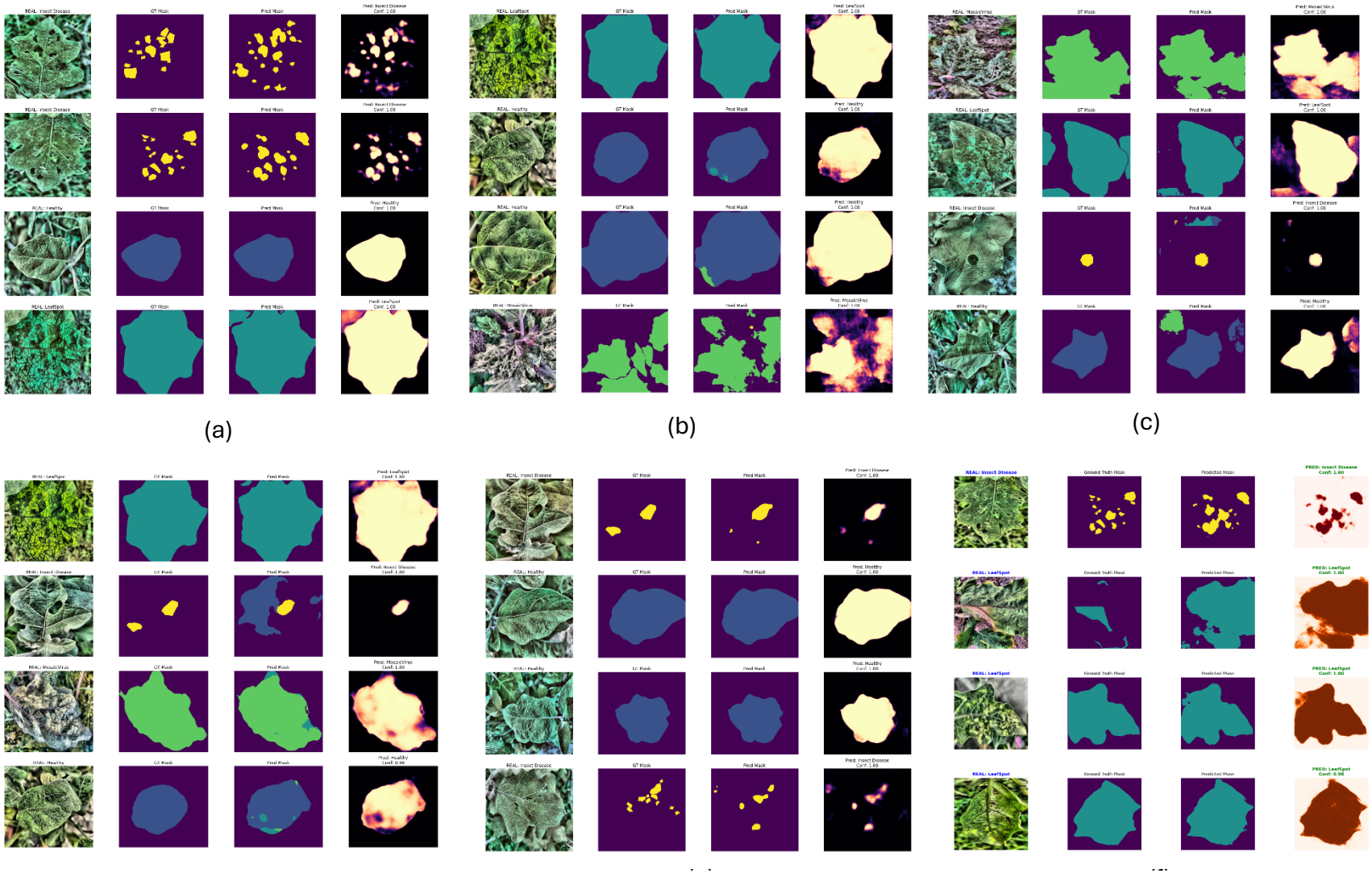


Fig. 12. Segmentation Samples for Eggplant Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table IX. Selected models observing predictive performance, hardware-aware metrics for TFLite-exported, and median and standard deviation for all models.

Crop	**Classification**			**Segmentation**		**#Epoch**	**Total Loss**	**Weigth (TFLite) [MB]**	**Parameters [M]**	**Inference [ms]**
	Accuracy [%]	**Recall [%]**	**F1-score [%]**	**Mean IoU [%]**	**Accuracy [%]**					
Tomate	98.21	98.21	98.21	66.03	90.2	22	0.4461	20.79	5.49	87.98
Potato	98.21	98.21	98.19	81.89	91.33	13	0.2255	13.1	3.32	68.72
Pepper	100	100	100	98.91	99.63	38	0.0234	9.88	2.63	59.81
Eggplant	97.5	97.5	97.49	81.69	93.51	21	0.5465	9.89	2.63	66.99
Median	***98.48***	***98.48***	***98.47***	***82.13***	***93.67***	***23.50***	***0.31***	***13.42***	***3.52***	***70.88***
Standar desviation	***0.92***	***0.92***	***0.93***	***11.63***	***3.64***	***9.07***	***0.20***	***4.46***	***1.17***	***10.43***

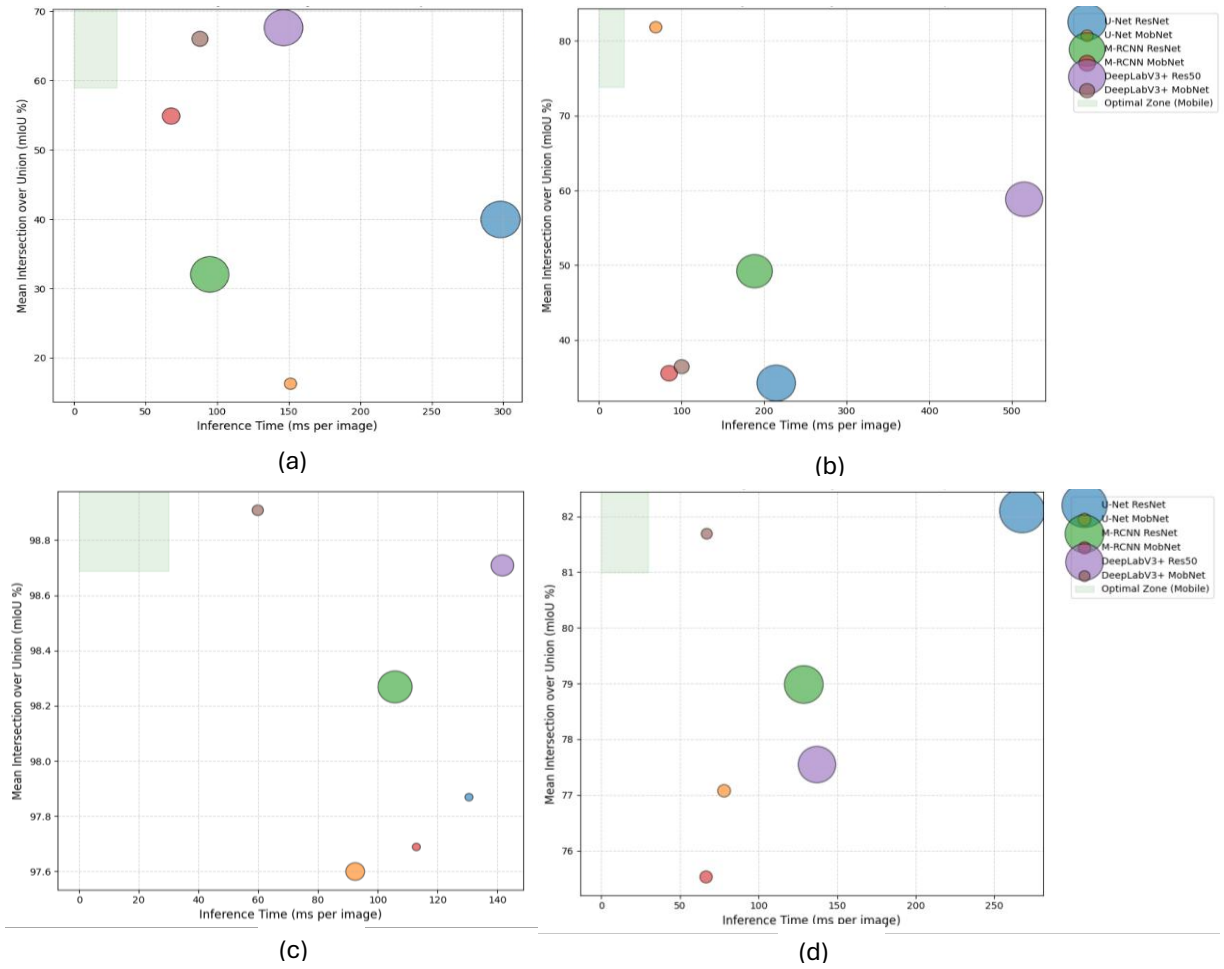


Fig. 13. Scatter plot. Trade-off analysis: Accuracy vs. Inference speed. (a) Tomato, (b) Potato, (c) Bell Pepper, (d) Eggplant.

Survey S1. User Satisfaction Survey

User Satisfaction Survey: SolanAPP Project

Institution: National Center of Animal and Plant Health (CENSA) / Agrarian University of Havana (UNAH)

Date: _____ / _____ / 2026

Participant Profile: [ ] Farmer, [ ] Agronomy Student/Scholar, [ ] Agronomist/Field Technician

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Instructions:

Please rate your experience with SolanAPP by marking the box that best represents your opinion for each statement, where:

- **1:** Strongly Disagree
- **2:** Disagree
- **3:** Neutral
- **4:** Agree
- **5:** Strongly Agree

Section 1: Usability and Interface

#	Statement	1	2	3	4	5
1.1	The application interface is intuitive and easy to navigate.	[ ]	[ ]	[ ]	[ ]	[ ]
1.2	The visual icons and menus are clear and easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
1.3	I felt confident using the app without constant technical assistance.	[ ]	[ ]	[ ]	[ ]	[ ]

Section 2: Functional Suitability and Transparency

#	Statement	1	2	3	4	5
2.1	Outlining the leaf spots in the picture helps me trust the diagnosis.	[ ]	[ ]	[ ]	[ ]	[ ]
2.2	The identification of the disease matches my visual observation.	[ ]	[ ]	[ ]	[ ]	[ ]
2.3	The diagnostic results provided by the app are accurate	[ ]	[ ]	[ ]	[ ]	[ ]

Section 3: LLM & Educational Value

#	Statement	1	2	3	4	5
3.1	The LLM-generated report successfully integrates both the computer vision diagnostic results and the user's qualitative field notes.	[ ]	[ ]	[ ]	[ ]	[ ]
3.2	The risk assessment provided by the LLM accurately correlates the disease's physical morphological features with the estimated percentage of affected area.	[ ]	[ ]	[ ]	[ ]	[ ]
3.3	The tailored action plans are clear and appropriately prioritize organic solutions for low/initial-severity levels over specific chemical treatments for critical cases.	[ ]	[ ]	[ ]	[ ]	[ ]
3.4	The practical management recommendations are useful and highly applicable to my daily field work or studies.	[ ]	[ ]	[ ]	[ ]	[ ]
3.5	The app helps me learn more about plant pathology while using it.	[ ]	[ ]	[ ]	[ ]	[ ]

Section 4: Reliability and Field Autonomy

#	Statement	1	2	3	4	5
4.1	The ability to perform offline diagnosis is essential for my work environment.	[ ]	[ ]	[ ]	[ ]	[ ]
4.2	I would use this app even if I have no internet access in the field.	[ ]	[ ]	[ ]	[ ]	[ ]
4.3	Sharing disease alerts by location (early warning) can help nearby farmers and the community	[ ]	[ ]	[ ]	[ ]	[ ]

Section 5: Qualitative Feedback (Optional)

What is the most useful feature of SolanAPP for you?

What improvements would you suggest for future versions?

Thank you for your participation.

Table X. User satisfaction survey results (n=25)

Sections	Total votes				
	5	**4**	**3**	**2**	**1**
1.1	17	5	3	0	0
1.2	18	6	1	0	0
1.3	10	13	1	1	0
Sum Section1	45	24	5	1	0
2.1	15	8	1	1	0
2.2	14	9	2	0	0
2.3	11	11	3	0	0
Sum Section2	40	28	6	1	0
3.1	13	9	2	1	0
3.2	13	8	4	0	0
3.3	11	13	1	0	0
3.4	16	9	0	0	0
3.5	19	4	2	0	0
Sum Section 3	72	43	9	1	0
4.1	17	7	1	0	0
4.2	14	8	3	0	0
4.3	19	4	2	0	0
Sum Section 4	50	19	6	0	0